

AIRLINE CUSTOMER SATISFACTION ANALYTICS

Understanding what drives passenger recommendation through ratings, review text, sentiment analysis, and machine learning

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Portfolio area	Customer Analytics NLP Machine Learning Business Interpretation
Dataset	21,788 airline reviews Review period: 2005-2023

KEY FINDING	Perceived value and operational reliability are the primary drivers of customer recommendation, not premium services or onboard extras.
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64.3%	96.3%	98.8%	#1
customers do not recommend the airline	best model accuracy	best model ROC-AUC	Value for Money predictor

01. Case Snapshot

This report reframes the airline customer analysis as a business analytics portfolio case study. It focuses on the business question, evidence generated, modelling results, decision value, and practical recommendations.

Case element	Summary
Business question	What drives airline customer recommendation behavior?
Dataset	21,788 airline customer reviews combining structured ratings and written feedback.
Analytical approach	Ratings analysis, text analytics, sentiment analysis, feature extraction, and classification modelling.
Best model result	Numerical Random Forest: 96.3% accuracy and 98.8% ROC-AUC.
Main insight	Perceived value and operational reliability are stronger recommendation drivers than onboard extras.

02. Business Question

Airlines receive large volumes of customer feedback, but recommendation behavior is not always explained by average rating scores alone. Customers may report mixed experiences, give inconsistent ratings, or describe dissatisfaction in written reviews that are not fully captured by numerical scores.

This project asks: how can airline review data be used to understand and predict customer recommendation behavior, and what service priorities can be identified from the analysis?

03. Data and Analytical Approach

The analysis used airline customer reviews containing structured service ratings and written passenger feedback. The work combined descriptive, text-based, and predictive methods in a single business analytics workflow.

- Data cleaning and review of missingness.
- Analysis of service ratings and recommendation outcomes.
- Text preprocessing and feature extraction using TF-IDF.
- Sentiment analysis using VADER and RoBERTa.
- Classification modelling using Logistic Regression, Linear SVM, and Random Forest.
- Model evaluation using accuracy, precision, recall, F1-score, and ROC-AUC.
- Interpretation of model outputs in business terms.

04. Key Insights

#	Insight	Business meaning
1	Value for Money is the strongest recommendation driver	Passengers rating value for money highly were far more likely to recommend. Recommendation rates stayed low at weak value ratings and rose sharply once customers perceived sufficient value.
2	Operational failures have an outsized negative effect	Refund problems, cancellations, delays, and poor customer support were strongly linked to non-recommendation. These issues appear more damaging than weaker performance in onboard extras.
3	Text adds diagnostic value	Structured ratings were highly predictive, but written review text helped explain why customers were dissatisfied.
4	Prediction needs business interpretation	The best model achieved strong predictive performance, but the value of the project comes from translating outputs into customer experience priorities.

05. Analytical Evidence

The analysis generated several evidence points that can be used by customer experience teams to understand recommendation behaviour and prioritise service improvement.

Evidence area	Business takeaway
Service-theme analysis	Refund, cancellation, and delay-related reviews showed much weaker recommendation behaviour than service-positive
Correlation analysis	Value for Money and Overall Rating were the strongest indicators of recommendation behaviour.
Model comparison	The Numerical Random Forest model achieved the strongest overall performance.
Language analysis	Positive reviews emphasised friendly staff, smooth flights, and comfortable seats. Negative reviews emphasised delays, cancellations, no refund, and poor customer service.

06. Model Performance

Machine learning models were trained to predict whether a passenger would recommend the airline. The strongest-performing model was the Numerical Random Forest model.

Model	Accuracy	Precision	Recall	ROC-AUC
Numerical Random Forest	96.3%	95.3%	94.3%	98.8%
Combined Linear SVM	96.2%	95.4%	93.8%	98.7%
Combined Random Forest	96.1%	95.3%	93.8%	98.8%
Numerical Logistic Regression	94.5%	92.9%	91.6%	97.9%
Text Linear SVM	94.1%	93.2%	90.0%	97.8%

INTERPRETATION

In business terms, the model results show that service ratings and customer review signals can strongly distinguish between customers who are likely to recommend and those who are not. Text features did not materially improve accuracy, but they added diagnostic insight into the specific service failures driving dissatisfaction.

07. Decision Value

This project does not claim that recommendation rates were improved in practice. Its value is analytical and decision-focused.

- Identify recommendation drivers.
- Detect dissatisfaction signals in written feedback.
- Prioritise service failure themes for operational review.
- Understand customer experience patterns beyond average scores.
- Support customer experience monitoring and service improvement decisions.

08. Recommendations

1. Prioritise perceived value

Value for Money was the strongest recommendation driver. Airlines should monitor perceived value as a core customer experience KPI and align pricing, service quality, and expectations more closely.

2. Improve disruption and service recovery

Cancellations, refund problems, delays, and poor support were strongly linked to non-recommendation. Faster resolution, proactive communication, and consistent compensation handling should be prioritised.

3. Use text analytics for complaint monitoring

Review text reveals dissatisfaction signals that ratings may miss. Sentiment and theme monitoring can help identify emerging service issues earlier.

4. Combine ratings and text for customer intelligence

Structured ratings are useful for prediction, while text is useful for explanation. Combining both gives a more complete view of customer experience.

09. Limitations and Next Steps

Limitations

- Online reviews may overrepresent dissatisfied passengers.
- Customer expectations changed across the long review period.
- Sentiment models can misread sarcasm or mixed sentiment.
- Predictive performance should not be treated as causal proof.

Next steps

- Recalibrate models using recent review data.
- Segment findings by airline, route, cabin class, or travel purpose.
- Test service recovery interventions against non-recommendation risk.
- Build a dashboard for ongoing complaint-theme monitoring.

10. Portfolio Reflection

This project strengthened my ability to connect machine learning outputs with business interpretation. The technical modelling was important, but the main value came from translating ratings, sentiment, and text patterns into customer experience priorities.

The project also reinforced an important business analytics lesson: high predictive accuracy is useful, but it must be paired with explainability, context, and practical recommendations for decision-makers.

**PORTFOLIO
TAKEAWAY**

Business analytics is not only about building accurate models. It is about converting analytical evidence into decisions that stakeholders can understand, trust, and act on.

Appendix: Tools and Methods

Tool / method	Use in project
Python	Data preparation, analysis, modelling, and visualisation.
pandas and NumPy	Data cleaning, transformation, and structured analysis.
TF-IDF	Text feature extraction for predictive modelling.
VADER and RoBERTa	Sentiment analysis of written review text.
scikit-learn	Classification modelling, model comparison, and evaluation.
Model evaluation	Accuracy, precision, recall, F1-score, and ROC-AUC.